Zhengzhong Ricky You

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Research Interests

- Exact algorithms for large-scale integer and combinatorial optimization, with an emphasis on branch-price-and-cut, decomposition, and extended formulations.
- Learning-augmented optimization, decision-focused learning, and reinforcement learning for combinatorial optimization, together with interfaces between operations research and machine learning.
- Applications of large language models to healthcare operations, including routing, scheduling, and resource allocation under uncertainty and fairness constraints.

Education

- **University of Florida**, M.S. in Operations Research, Industrial & Systems Engineering; GPA: 3.95/4.00 **2025**
- **Nankai University**, B.S. in Chemistry; GPA: 3.67/4.00 **2021**

Publications

[S1] *Learned Pairwise Deep Dual-Optimal Inequalities for Stabilizing Column Generation*.
Zhengzhong Ricky You, Bo Tang, Haoran Liu, Baichuan Mo. Under review at **INFORMS Journal on Computing**, 2026.

[S2] *Bulk Service Queueing for Transit Resilience under Short Random Service Suspensions*.
Baichuan Mo, Li Jin, **Zhengzhong Ricky You**, Zuo-Jun Max Shen, Haris N. Koutsopoulos, Jinhua Zhao. Under review at **Transportation Science**, 2026.

[C1] *Draft-and-Audit Reinforcement Learning for Optimization Modeling*.
Zeping Min, Weihang Xu, **Zhengzhong Ricky You**, Wotao Yin, Xinshang Wang. **ICML 2026**.

[J1] *RouteOpt: An Open-Source Modular Exact Solver for Vehicle Routing Problems*.
Zhengzhong Ricky You, Yu Yang. **INFORMS Journal on Computing**, 2026.

[J2] *Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly*.
Zhengzhong Ricky You, Yu Yang, Xinshang Wang, Wotao Yin. **Operations Research**, 2026.

Research Experience

- **TikTok Inc.** **2026**
  Machine Learning Engineer Intern, E-commerce Supply Chain & Logistics *San Jose, CA*
  - Conducted research on inventory control for e-commerce supply chains under the mentorship of Baichuan Mo, Yikang Hua, and Zhou Ye.
- **MIT-UF-NEU Joint Summer Research Camp** **2026**
  Selected Participant, MIT JTL-Transit Lab program
  - Studied large-scale inventory placement under demand uncertainty with mentorship from Baichuan Mo.
- **University of Kentucky** **2026**
  Visiting Scholar, Department of Statistics
  - Hosted by Chenglong Ye; conducted research on online learning and branch-and-bound algorithms.
- **DAMO Academy, Alibaba Group** **2025**
  Research Intern *Bellevue, WA*
  - Studied solver parameter tuning with large language models under the mentorship of Xinshang Wang and Wotao Yin. The resulting configurations improved average benchmark performance relative to solver defaults, whereas tested black-box baselines generally did not.

Selected Relevant Courses

ISE Ph.D. level (GPA: 4.00): Discrete Optimization; Linear Programming and Network Optimization; Stochastic Modeling & Analysis; Advanced Topics in Continuous Optimization. **Mathematics Ph.D. level (GPA: 4.00)**: Analysis I; Numerical Linear Algebra; Numerical Analysis; Numerical Optimization. **CS Ph.D. level (GPA: 4.00)**: Machine Learning.

Research Software

- **TorchDCM** — **2026**
  Co-developer, Python — *Open source*
  - Co-developed a unified PyTorch-native package for discrete choice modeling. The package implements vectorized likelihoods and automatic differentiation across multiple logit model families and provides standard econometric outputs.
- **RouteOpt** — **2021**
  Lead Developer, 20k+ LOC in C++ — *Open source*
  - Designed and implemented an open-source exact solver for the CVRP and CVRPTW. The solver uses modular branching, cutting-plane, and variable-reduction components, learning-augmented branching, and reproducible experiment scripts.
  - On benchmark instances, RouteOpt nearly doubled the speed of VRPSolver and proved optimality for four previously open instances; one required approximately 1.5 years of single-core time.

Conference & Workshop Presentations

- *L-DDOIs: Learning-Based Deep Dual-Optimal Inequalities for Stabilizing Column Generation*, INFORMS Transportation Science and Logistics Conference, MIT Sloan — **2026**
- *Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly*, invited talk, UK AI and ML Symposium and Nontechnical Workshop — **2025**
- *L-DDOIs: Learning-Based Deep Dual-Optimal Inequalities for Stabilizing Column Generation*, INFORMS Annual Meeting — **2025**
- *Accelerating Dynamic Programming via Dual Selection for Column-Generation Frameworks*, INFORMS Optimization Society Conference — **2024**
- *RouteOpt: A Scalable Advanced Optimization Tool for VRPs*, INFORMS Annual Meeting — **2024**
- *Two-Stage Learning to Branch in Branch-Price-and-Cut for Exact VRP*, INFORMS Annual Meeting — **2023**
- *Learning to Branch with Column Generation*, INFORMS Annual Meeting — **2022**

Awards & Honors

- Harold D. Haldeman, Jr. Graduate Fellowship, University of Florida — **2024**
- INFORMS Optimization Society Conference Travel Support — **2024**
- Academic Excellence Scholarship, Nankai University — **2020**
- Innovation and Entrepreneurship Scholarship, Nankai University — **2019**
- First Prize, Tianjin Theoretical Chemistry Competition — **2019**
- First Prize, Tianjin Mathematics Competition — **2018**

Teaching & Mentoring

- **Sole Instructor**, ESI 3327C Matrix & Numerical Methods in Systems Engineering, University of Florida; taught linear algebra, numerical methods, and introductory optimization to 52 undergraduates — **2024**
- **Outreach Mentor**, UF Student Science Training Program; Ken Cheng, Columbia University (computer science), and Shrey Gupta, University of Oxford (mathematics)

Professional Service

Journal referee for *INFORMS Journal on Computing*, *INFORMS Journal on Optimization*, *Transportation Science*, *Transportation Research Part B*, *IISE Transactions*, *Computers & Operations Research*, and *OR Spectrum*.

Skills

Methods: branch-price-and-cut, column generation, dual stabilization, cutting planes, learning-augmented optimization, stochastic and robust optimization. **Programming**: C++, Python including PyTorch and reinforcement learning libraries, MATLAB, Shell, and Git. **Software**: RouteOpt, TorchDCM, Gurobi, CPLEX, SCIP, and reproducible experiment pipelines.

Updated 2026